

Task 10: Decision Trees and Random Forests - Interview Questions & Answers

Q: Tree vs Forest?

A Decision Tree is a single tree model, while a Random Forest is a collection of many decision trees whose predictions are combined.

Q: Gini Impurity vs Information Gain?

Gini measures node impurity. Information Gain measures reduction in entropy after a split.

Q: How does Random Forest handle overfitting?

It uses multiple trees trained on different samples and combines their predictions, reducing variance.

Q: What is Bagging?

Bootstrap Aggregating: training multiple models on random samples and combining predictions.

Q: Why is Random Forest called an Ensemble Method?

Because it combines multiple models (decision trees) into one stronger model.

Q: What is Feature Importance?

A score showing how much each feature contributes to predictions.

Q: What is entropy in Decision Trees?

Entropy measures disorder or uncertainty in data.

Q: What is pruning?

Removing unnecessary branches from a tree to reduce overfitting.

Q: Advantages and disadvantages of Decision Trees?

Easy to understand but can overfit and be unstable.

Q: Advantages and disadvantages of Random Forest?

High accuracy and robustness but less interpretable and slower.

Q: What are hyperparameters in Random Forest?

Settings like `n_estimators`, `max_depth`, and `min_samples_split`.

Q: What is bootstrap sampling?

Sampling with replacement from the training dataset.

Q: How does majority voting work?

The class predicted by most trees becomes the final prediction.

Q: Difference between Bagging and Boosting?

Bagging builds models independently; Boosting builds models sequentially.

Q: Can Random Forest handle missing values?

Some implementations can, but preprocessing is generally recommended.

Q: What evaluation metrics are used for classification?

Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Q: What is a classification report?

A summary of Precision, Recall, F1-Score, and Support.

Q: What is overfitting and underfitting?

Overfitting learns noise; underfitting fails to capture patterns.

Q: Why is feature scaling usually not required for trees?

Tree splits depend on feature ordering, not magnitude.

Q: Real-world applications?

Medical diagnosis, fraud detection, credit scoring, customer churn prediction.